

6(a)	one mark each correct row	3												
	<table> <tr> <th>feature</th><th>mRNA</th><th>DNA</th></tr> <tr> <td>names of four bases</td><td><u>adenine</u>, <u>cytosine</u>, <u>guanine</u>, <u>uracil</u></td><td><u>adenine</u>, <u>cytosine</u>, <u>guanine</u>, <u>thymine</u> ;</td></tr> <tr> <td>name of pentose sugar present</td><td>ribose</td><td>deoxyribose ;</td></tr> <tr> <td>number of strands</td><td>1 ; I single</td><td>2 ; I double</td></tr> </table>	feature	mRNA	DNA	names of four bases	<u>adenine</u> , <u>cytosine</u> , <u>guanine</u> , <u>uracil</u>	<u>adenine</u> , <u>cytosine</u> , <u>guanine</u> , <u>thymine</u> ;	name of pentose sugar present	ribose	deoxyribose ;	number of strands	1 ; I single	2 ; I double	
feature	mRNA	DNA												
names of four bases	<u>adenine</u> , <u>cytosine</u> , <u>guanine</u> , <u>uracil</u>	<u>adenine</u> , <u>cytosine</u> , <u>guanine</u> , <u>thymine</u> ;												
name of pentose sugar present	ribose	deoxyribose ;												
number of strands	1 ; I single	2 ; I double												
6(b)(i)	transcription ;	1												
6(b)(ii)	<i>can award either mp1 with mp2 ,or mp3 with mp4</i> 1 removal of introns (from primary transcript / RNA) ; 2 exons joined together ; or 3 splicing ; 4 of, primary transcript / RNA ;	2												

Question	Answer	Marks
6(c)(i)	any four from: 1 <i>ref. to</i> amino acid activation / charging amino acids before attachment to tRNA / AW ; A aminoacylation 2 tRNA attached to, specific amino acid / glycine ; R 'picks up' 3 <i>idea that</i> anticodon / CCG, identifies specific amino acid ; 4 tRNA transports / AW, amino acid to ribosome ; 5 <i>ref. to</i> (tRNA) entering the A (aminoacyl) site in ribosome ; 6 tRNA, anticodon / CCG, binds to mRNA, codon / GGC ; A correct <i>ref. to</i>, complementary pairing / base pairing / H bond formation <i>as AW for binding</i> 7 (specific) amino acid / glycine, in correct position in the polypeptide chain / AW ; A helps to form / AW, primary structure of cotransporter protein 8 AVP ; e.g. tRNA, recycled / reused forms aminoacyl tRNA / amino acid-tRNA complex tRNA-ligase / aminoacyl-tRNA synthetase	4
6(c)(ii)	any three from: 1 movement of sucrose with, protons / H⁺ ions, into companion cells ; A correct <i>ref. to</i> (other named) assimilates 2 (sucrose moved) from, apoplast / cell wall / mesophyll cell ; 3 sucrose moves against its concentration gradient (into companion cell) ; 4 (needs cotransporter protein because) sucrose is polar so cannot pass through membrane ; 5 maintain concentration gradient for sucrose between companion cell and sieve tube (element) ;	3